10. SUMMARY SCOPE OF WORK & SERVICES [TB-5, ITEM 1 AND 3]

This section provided the details of engineering and associated services to be carried out by ENGINEER during the Select Stage Engineering (SELECT Stage Engineering and Select Design) of the project, which shall include any services necessary to fulfil the objectives and requirements of the Project, even if not specifically mentioned herein.

The summary of the SERVICES to be provided by the ENGINEER is broadly as follows:

1. Project Management Services including planning, project control and administration.

2. Interface management (within and outside COMPANY).

3. Preparation of QA/QC plans and procedures.

4. Perform the SERVICES in accordance with applicable local regulations, international design codes and standards, Industry Best Practice and COMPANY procedures, standards, and specifications.

5. Data collection through COMPANY systems and from other ENGINEERs or site as necessary.

6. Preparation of engineering documentation related to the SERVICES as defined in the ITT SOW

7. Site visit to existing facilities for data collection including review and identify any discrepancies of as built drawings of applicable existing facilities. Discussion with COMPANY project and operations personnel at site and Abu Dhabi offices.

8. Review existing documentation and philosophies, obtain from COMPANY the latest production profile and final number of wells, verify& confirm if any changes in fluid characteristics such as H2S content, high GOR etc., verify and revise/generate HYSYS Simulations to establish revised properties, prepare process data sheets for all new facilities including Choke Valves and MOVs.

9. ENGINEER, during SELECT stage, shall either close all the open points from the workshops conducted during ASSESS stage or carry them to DEFINE stage in consultation with COMPANY. the study shall be agreed with COMPANY prior to commencement of the study.

10. Multidiscipline Technical assessments ENGINEER shall perform multidiscipline Study to assess technical feasibility and constraints of the existing facilities as well as to perform technical definition of the strategies identified. Technical assessments including third-party studies shall be integrated into the technical definition of the each of the identified strategies. It includes preparing multi-discipline deliverables including BFDs, equipment lists, pipeline routing, high level layout and footprints for the new facilities, estimation of structural weights, identification of requirements for new pipelines and cables, preliminary load list, artificial island and long-term jetty BOQs etc.

11. Centre of Excellence

ENGINEER during the course of CONCEPT SELECT shall require engaging its Centre of Excellence for conducting specialized/high tech studies such as autonomous operations (Rock to Stock without human intervention), Digital Oil Fields, Decarbonization, BAT, etc. ENGINEER shall engage and obtain the necessary assistance and expertise from his other design centres with consent of COMPANY, if required in the execution of the SERVICES. Such assistance shall be at no additional cost to the

COMPANY. Adopt autonomous lean operations focusing on optimizing accommodation and operational personnel numbers.

12. Synergy Development AOF and AON

ENGINEER shall study the Synergy between offshore and onshore where the production is routed to Onshore Facilities for further processing. A new processing facility will be designed and executed on the onshore. This will include studying support facilities for the offshore operation from onshore (i.e. maintenance workshops, warehousing, accommodation, etc)

As part of synergy with UAD/ULL/Al Nouf, the following aspects shall be included in the Optimization Study:

- New pipeline to Al Nouf shall be utilized for ULL as evacuation route for their business continuity (through existing 12" pipeline between UA CPC and ULSC)
- New Pipeline to Al Nouf to be evaluated for multi-scenarios of production schemes, such as full well stream, gas only, or 2 phases (oil & gas), based on production profile provided
- Unmanned and Autonomous operation philosophy for pre-investment facilities to be managed from Al Nouf or UAD CPC

ENGINEER shall study and verify all the data and studies available during the ASSESS stage related to the synergy requirement, including but not limited:

- Feasibility Report prepared by ADNOC Onshore/Offshore
- Feasibility Report (ASSESS STUDY REPORT) prepared during Assess

13. Sector Analysis

ENGINEER shall perform sector analysis workshop for selecting the location of new structures. The existing Helideck operation shall not be hindered by installation of new facilities, and if inevitable appropriate mitigation measures shall be proposed to COMPANY for review and acceptance. To locate the new structures suitable clearance shall be maintained from existing subsea cables and other existing items. The new Deck extensions/bridge connected towers proposed shall be reviewed and ensure that Helidecks are in line with CAAP requirements. COMPANY will involve Aviation technical authority in the Sector Analysis workshops to raise any areas of concern before finalization of locations.

15. Optimization Studies

ENGINEER shall study following optimization ideas in addition to above Selected Strategies:

a. Convert existing 14" MOL pipeline to use as Gas Export pipeline & install a new MOL pipeline for 60 MBD oil transfer either UZ ZCC, UZ Zirku or Al Nouf.
b. Evaluate necessary measure and operation mode to accommodate higher flowrates (i.e. 45 or 60 MBD) to UZ ZCC through existing 14" MOL pipeline, e.g. upgrade of MOL Pump and Discharge Manifold, application of DRA (Drag Reducing Agents), etc.
c. Convert Associated Gas to Liquid (GTL) suitable to our process conditions located offshore and transfer liquid to nearby facilities for further processing or separately handling of liquid

products. Further, this option shall be techno economically compared with New Gas pipeline system required for Gas export.

d. As Base case New CPC platforms to cater total 60 MBD crude (Avg Annual).
e. As optimization option, utilize full capacity of existing Central Processing Complexes and build new processing facilitates only for additional flows beyond the available capacity of existing facilities to meet the Oil, Gas, produced water and Water injection production target.
f. Al Nouf facilities sizing for multiple production scenarios such as 32, 40-50 32-60 [TB-5, ITEM 1], 60 and 60+ MBD. Also, evaluate facility requirements for production scenarios where 32 MBD oil production will be transferred to UZ ZCC and the balance transfer to Al Nouf Facilities.
g. Evaluate techno-economically optimum solution to achieve flare gas recovery requirement with the existing CPC facilities and latest operational scheme in Sustain Study (32MBD) such as conversion of temporal UAD Gas Spike compressor into permanent one, new gas export pipeline etc.
h. Evaluate feasibility of utilization of Existing WHTs (i.e. WHT-3, 17, 27, 31, and 32) including Well Slot Extension, instead of New Well Head Tower.
i. Consider new WHTs with more than 12 Well Slots.
j. In case of new jacket requirement, ENGINEER shall study an alternative option to utilize retiring Jack-up Rig/Barge to have new topside facilities (utilities, New GTG (if applicable), WI Model and accommodation etc..) with economics evaluation based on market survey (for Sustain and Mid-scale Scenario only).
k. Power import from ULL to UAD or install new GTG at CPC to meet the power demand for Sustain 32 and Mid-scale Study (40-50 32-60 [TB-5, ITEM 1] MBD) facilities.
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m. demand for Sustain 32 and Mid-scale Study (40-50 32-60 [TB-5, ITEM 1] MBD) facilities.
n. Additionally, captive power generation using new Gas Turbine Generators Offshore shall be studied as an alternative to Grid /onshore power.
o. Unmanned and Autonomous operation philosophy for pre-investment facilities to be managed from Al Nouf or UAD CPC.
p. Study to explore the options to minimize the GTG emissions to meet COMPANY decarbonization objective.
q. Gas Export Optimization Study: Gas separated at UAD is proposed to be exported to DAS Island via the 34" NMGL at Zakum West Super Complex (ZWSC), following compression and dehydration as per ASSESS study outcome. Moreover as part of optimization studies, bidders shall consider an option of common gas export pipeline system for both the Umm Lulu & UAD fields, either from ZWSC to DAS Island or an alternative route options i.e. Al Nouf via ULL or UAD, DAS via ULL to Zirku, etc. The successful bidder will be provided UAD & ULL Excess gas production profile. Bidders shall study the requirement of new pipelines & associated facilities, develop adequacy assessment of existing facilities and propose best techno-economic Gas export option based on the study's assessment findings.[TB-5, ITEM 3]